

JKBOSE 2026

CLASS 11TH

BIOLOGICAL CLASSIFICATION

LECTURE-08

BOTANY

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Topics To Covered

- 1** Classification of Fungi ✓

BASIS OF CLASSIFICATION OF FUNGI

On the basis of

- ✓ a. Morphology of fungi
- ✓ (b) Mode of spore formation.
- ✓ (c) Mode of fruiting body formation

Related with Sexual reproduction.

4 classes of Fungi :

- ①. Phycomycetes ✓
- ②. Ascomycetes ✓
- ③. Basidiomycetes ✓
- ④. Deuteromycetes... ✓

Note: Different types of Sexual Spores → Oospores / Zygosporos ✓
→ Ascospores. ✓
→ Basidiospores. ✓

PHYCOMYCETES

- lower fungi.
- AKA Algal fungi

①. Habitat and Nutrition:

Aquatic, Decaying wood, moist & damp places, obligate parasites on plants.

②. Mycellium: Aseptate and Coenocytic (multinucleated)

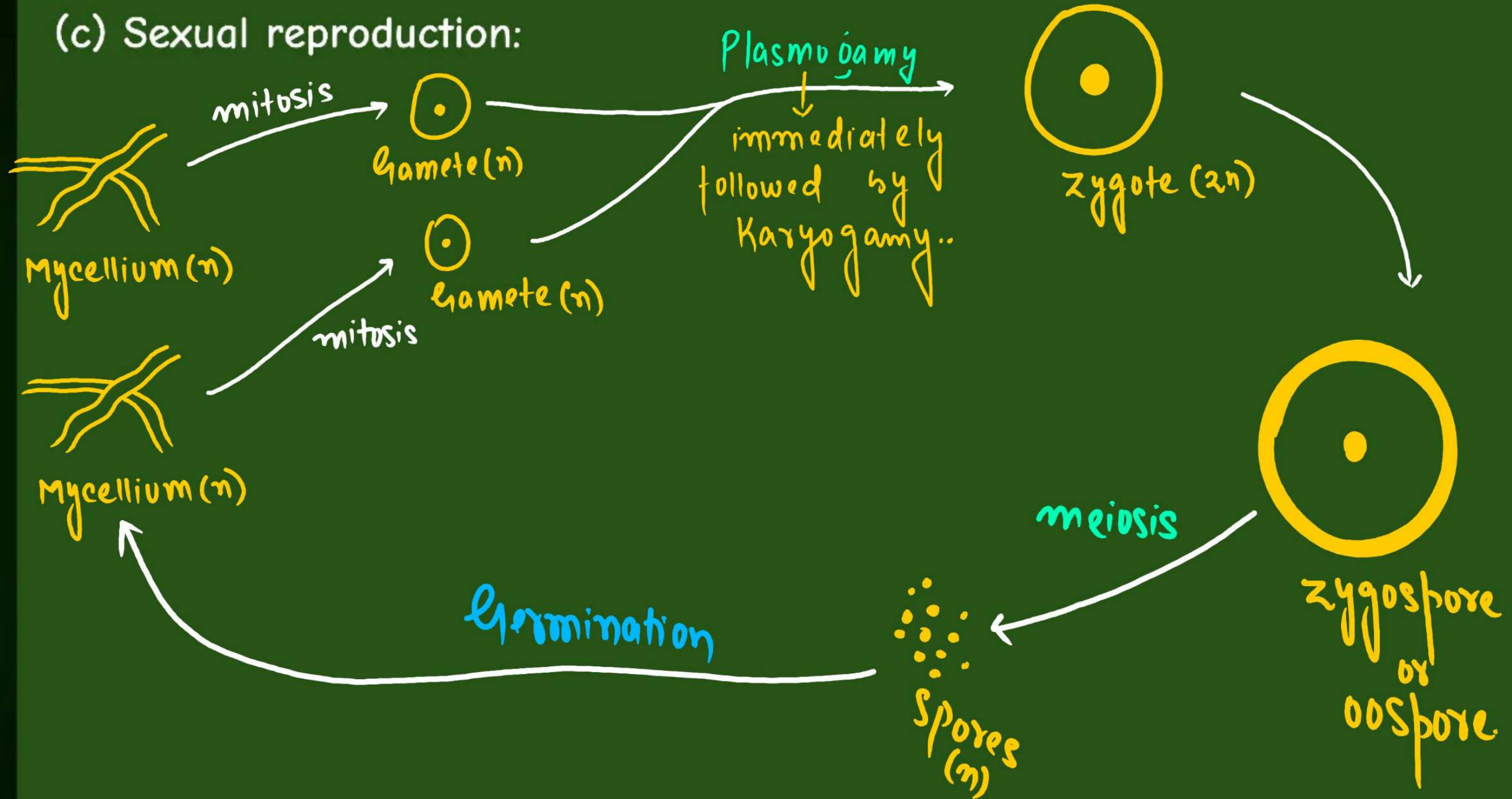
③. Reproduction:

a) Vegetative reproduction: By Fragmentation.

b) Asexual reproduction: By 

- Zoospore (motile)
- Aplanospore (non-motile)

(c) Sexual reproduction:



Sexual spore: $+nt.$ (zygospore / oospore)

Fruiting body: $-nt.$

Sex gametes: $+nt$

Both gametes are similar (isogamy)

Both gametes are similar in size & motile.


Both gametes are similar in size and non-motile.


Both gametes are dissimilar (heterogamy)

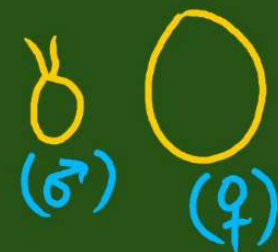
Anisogamy

Both gametes are flagellated but dissimilar in size



Oogamy

• one gamete is flagellated & small.
 • 2nd gamete is non-flagellated & large sized



Dikaryophase: $(-nt)$.

Examples of Phycomycetes: *Mucor* (Dung mould).
Rhizopus (Bread mould).
Albugo *candida*.

ASCOMYCETES

• Commonly kld as Sac fungi

Habitat and Nutrition:

• on decaying organic matter, decomposers
• Coprophilous (grows on dung), Saprophytic, Parasitic.

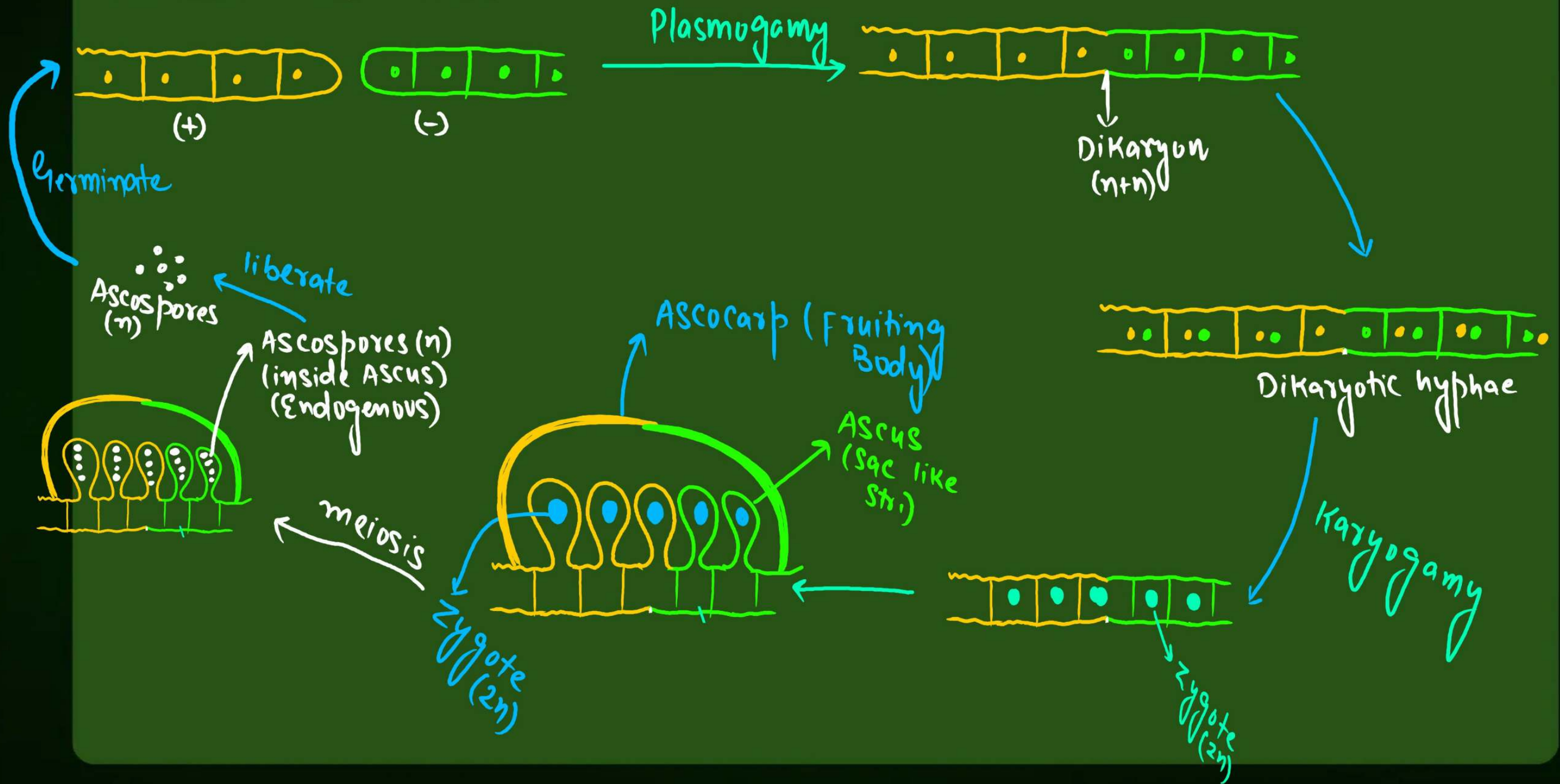
Mycellium: Branched and Septate mycellium.

Reptoduction:

a) Vegetative reproduction: Fragmentation, Budding, Fission

b) Asexual reproduction: By conidia

(c) Sexual reproduction:



Sexual spore: Ascospore (formed endogenously inside Ascus).

Fruiting body: Ascocarp (Aggregation of Ascus).

Sex organs: +nt

Sex gametes: +nt

Dikaryophase: +nt

Examples of Ascomycetes:

Yeast, Penicillium, Aspergillus, Claviceps,
Morels & truffles, Neurospora (Extensively used in biochemical & genetic work).

BASIDIOMYCETES

• Commonly kld as Club fungi.

Habitat and Nutrition:

Soil, wooden logs, tree stumps
• parasites on living plant bodies (eg. Rusts, Smuts).

Mycellium: Branched and Septate

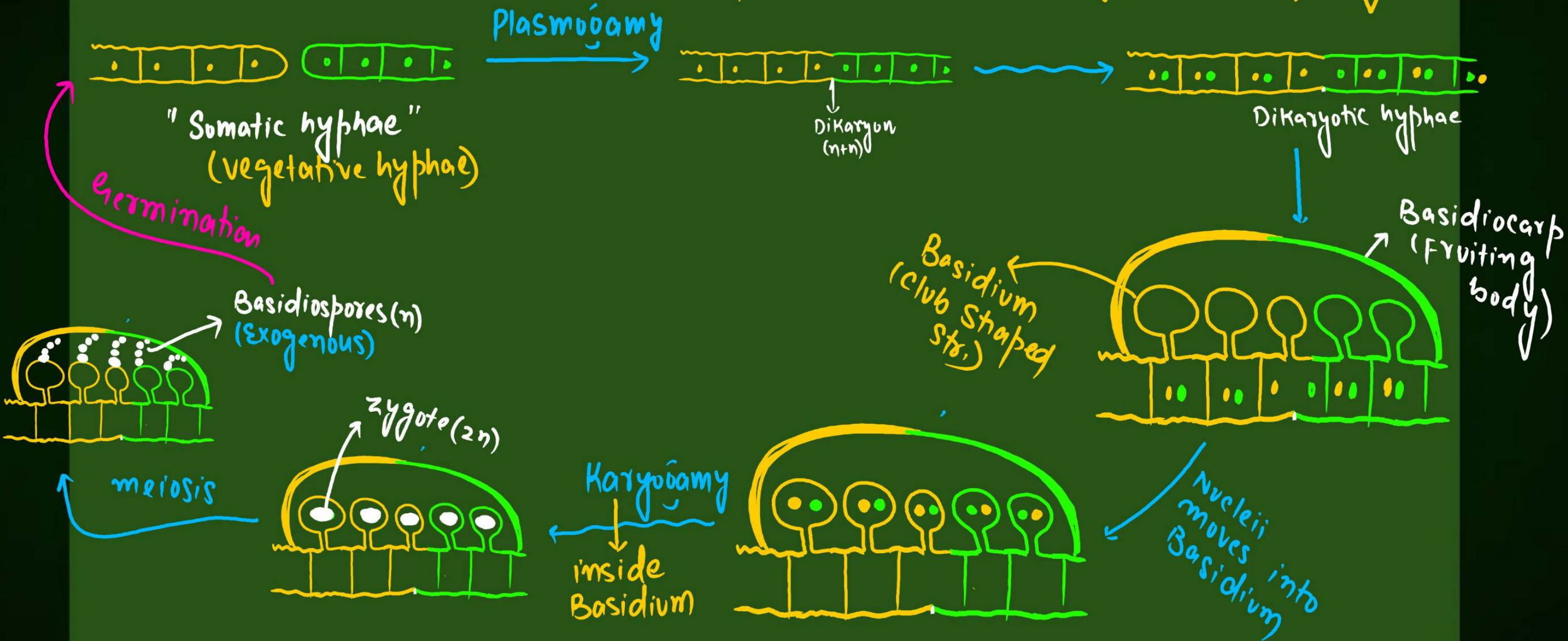
Reptoduction:

a) Vegetative reproduction: Fragmentation.

b) Asexual reproduction: Absent.

(c) Sexual reproduction:

- Sex organs - nt.
- No sex gametes.
- Sexual reproduction occurs by "Somatogamy"



Sexual spore: Basidiospore (exogenously at tip of Basidium)
= 04 Basidiospores are formed at tip of basidium.

Fruiting body: Basidiocarp,
"Aggregation of basidia"

Sex organs: -nt

Sex gametes: -nt

Dikaryophase: +nt

Examples of Basidiomycetes:

Agaricus, Bracket fungi, Puff balls, Ustilago (Smut in wheat),
Puccinia (Rust fungi), Toadstools (poisonous).

DEUTEROMYCETES

• AKA *fungi imperfecti*.

Habitat and Nutrition:

- Grows on *litter*,
- *Maximum* are *decomposers* (help in Nutrient cycling).
- Some — *Saprophytic* / *parasitic*.

Mycellium: *Septate and Branched*

Reptoduction:

a) Vegetative reproduction: *Fragmentation*.

b) Asexual reproduction: *conidia*.

(c) Sexual reproduction: "Absent"
ie, why deuteromycetes are kld as imperfect fungi.

NOTE: when sexual stages of these fungi were known, they are moved into the class - they rightly belong to.

Sexual spore: -nt

Fruiting body: -nt

Sex organs: -nt

Sex gametes: -nt

Dikaryophase: -nt

Examples of Deuteromycetes: Alternaria,
Trichoderma,
Colletotrichum...

QUESTIONS

Which among the following is not a prokaryote?

[NEET-2018]

- ☒ 1 Saccharomyces
- ☐ 2 Mycobacterium
- ☐ 3 Nostoc
- ☐ 4 Oscillatoria

QUESTIONS

H/W

After karyogamy followed by meiosis, spores produced exogenously in:

[NEET-2018]

- 1 Neurospora
- 2 Alternaria
- 3 Agaricus
- 4 Saccharomyces

QUESTIONS

H/W

Assertion: In Ascomycetes and Basidiomycetes well distinct Dikaryophase present during sexual life.

Reason: During sexual reproduction of Ascomycetes and Basidiomycetes plasmogamy is immediately followed by karyogamy.

- 1 If both assertion and reason are true and reason is the correct explanation of assertion
- 2 If both assertion and reason are true but reason is not the correct explanation of assertion
- 3 If assertion is true but reason is false
- 4 If both assertion and reason are false

QUESTIONS

H/w

Which one of the following is not a criterion for classification of fungi?

[NEET-2024]

- 1 Mode of spore formation
- 2 Fruiting body
- 3 Morphology of mycelium
- 4 Mode of nutrition

QUESTIONS

H/W

Match List-I with List-II:

Choose the correct answer from the options given below:

[NEET-2024]

- 1 A-III, B-II, C-I, D-IV
- 2 A-IV, B-III, C-II, D-I
- 3 A-III, B-II, C-IV, D-I
- 4 A-I, B-III, C-II, D-IV

List-I		List-II	
A.	<i>Rhizopus</i>	I.	Mushroom
B.	<i>Ustilago</i>	II.	Smut fungus
C.	<i>Puccinia</i>	III.	Bread mould
D.	<i>Agaricus</i>	IV.	Rust fungus

Thank
You